

SOFTWARE ENGINEERING – PROJECT PROPOSAL

Eventra

A Web-Based Event Discovery and Management Platform

Prepared by

Team StackNova

Software Engineering and Information System Design Lab

Course Code: CSE-3104

Supervised by

Md Samsuddoha

Assistant Professor

Department of Computer Science & Engineering

University of Barishal

Submission Date: 17 May 2026

1. Team Formation

Eventra is being developed by **Team StackNova**, a six-member student team from the Department of Computer Science and Engineering. The team was formed based on complementary technical, analytical, design, testing, and documentation skills. Each member has a defined responsibility so that the project can be developed in an organized way without unnecessary task overlap.

Team Name: StackNova

Project Name: Eventra

#	Name	Student ID	Role
01	MD. Tanvir Ahmed Rana	230102016	Team Leader: Project coordination, task distribution, requirement analysis, documentation, SDLC planning, Git collaboration, HTML, CSS, PHP, and overall project supervision.
02	MD. Readul Haque Sohan	230102015	Backend Developer: Laravel back-end development, authentication, role-based access, event management logic, registration workflow, and API/module integration.
03	Rabea Amrin Linat	230102047	Database Administrator: Database schema design, ERD preparation, normalization, MySQL table relationship design, query optimization, and data security planning.
04	Niloy Deb	230102030	UI/UX Designer and Frontend Developer: Interface design, responsive layout, event card design, dashboard UI, HTML, CSS, JavaScript, and Bootstrap/Tailwind implementation.
05	Syeda Fazilatunnassa	230102011	SQA Engineer: Test planning, functional testing, system testing, bug tracking, validation of registration, QR attendance, certificate, and dashboard modules.
06	Sarmin Sultana Ritu	230102006	Business Analyst and Documentation Coordinator: Problem analysis, market and user research, feature planning, report writing, budget estimation, timeline preparation, and final documentation support.

Table 1: Team StackNova – Member Roles

2. SWOT Analysis of Team

Before starting development, Team StackNova conducted a SWOT analysis to understand the team's strengths, weaknesses, opportunities, and threats. This analysis helps the team plan the project realistically and identify areas that require extra attention during development.

Strengths

- The team has balanced skills covering frontend development, backend development, database design, testing, documentation, and project coordination.
- The project idea is practical and connected to real event management problems faced by students, communities, clubs, institutions, local groups, and independent organizers.
- Each member has a clear responsibility, which reduces task overlap and improves project organization.
- The team is familiar with important technologies such as HTML, CSS, JavaScript, PHP, Laravel, MySQL, Git, and responsive design.
- The initial project scope is realistic for a semester project while still keeping a scalable future vision.

Weaknesses

- The development time is limited because of academic coursework, examinations, and other responsibilities.
- QR-based attendance, certificate generation, and notification features require careful implementation and testing.
- Advanced features such as paid ticketing, AI recommendation, mobile apps, and large-scale deployment are outside the first version.
- The project depends on consistent communication and collaboration among all team members.

Opportunities

- Eventra can serve different user groups, including students, clubs, communities, institutions, local organizations, and independent event organizers.
- The platform can start with city-based and community-based events and later expand to different regions.
- A web-based system can be accessed from different devices without requiring separate installation.
- Future expansion is possible through mobile applications, paid events, AI-based recommendations, organizer subscriptions, and map-based discovery.

Threats

- Users may initially prefer familiar tools such as Facebook events, Google Forms, spreadsheets, or messaging groups.
- Established event platforms may already have strong user bases and advanced features.
- Security concerns may arise regarding user data, QR attendance records, certificate verification, and account authentication.
- Server downtime or poor internet connectivity may affect event check-in and real-time data operations.

3. Idea Generation

The team explored several problem areas before selecting event discovery and management as the project domain. During initial discussions, it became clear that many events are still managed through scattered tools. This problem exists among students, communities, clubs, training groups, professional organizations, social groups, institutions, and independent event organizers.

3.1 Domain and Problem Identification

The selected domain is **Event Technology**, with a focus on digital event discovery, event publishing, registration, attendance management, and post-event reporting. Events can be educational, professional, social, cultural, community-based, or skill-development focused. Examples include seminars, workshops, meetups, competitions, training sessions, community gatherings, club programs, and institutional events.

Currently, many small and medium-scale organizers depend on social media announcements, Google Forms, spreadsheets, manual check-in lists, and informal messaging. These tools may work for very small events, but they become difficult to manage when participant numbers increase or when organizers need attendance records, feedback, certificates, and reports. Therefore, there is a need for a simple, organized, and scalable event discovery and management platform.

3.2 Why Eventra Was Selected

- **Feasibility:** Eventra can be realistically developed within an academic semester using HTML, CSS, JavaScript, PHP Laravel, MySQL, and related web technologies.
- **Impact:** It can reduce manual work, improve event discovery, support participant management, and help organizers manage events more professionally.
- **Innovation:** The project combines event publishing, location-based discovery, organizer approval, QR attendance, feedback, certificate generation, notifications, and dashboards in one platform.

- **Scope:** The proposed system can be built as a working web-based prototype while keeping future options for mobile apps, online ticketing, and wider regional expansion.
- **Learning Value:** The project covers requirement analysis, UI/UX design, database design, backend development, testing, deployment, and documentation.

4. Introduction and Overview

Eventra is a modern web-based event discovery and management platform designed to connect event organizers and participants through one organized digital system. The platform allows individuals, communities, clubs, institutions, local organizations, professional groups, and independent organizers to create, publish, manage, and promote events from different locations. At the same time, participants can discover suitable events based on location, category, date, and interest.

The main idea of Eventra is to make event discovery and participation easier for both organizers and attendees. An organizer may publish a workshop in Barishal, a seminar in Dhaka, a community meetup, a training session, a cultural program, a club activity, or an institutional event. Interested participants can browse available events, view event details, register for participation, and receive confirmation after organizer approval. This makes the platform useful for educational, professional, social, cultural, and community-based events.

Eventra also supports the management side of events. Organizers can manage participant registrations, approve or reject requests, track attendance using QR codes, collect post-event feedback, generate certificates for eligible attendees, and monitor event activity through a dashboard. Admin users can manage users, event categories, published events, and overall platform activity.

Although Eventra has a broad scalable vision, the first version will remain realistic for an academic semester project. The initial version will focus on core features such as user registration, event publishing, participant registration, search/filter options, approval workflow, QR attendance, feedback, certificates, notifications, and dashboards. The goal is to build a practical academic prototype that demonstrates how a modern event platform can reduce manual work, improve event visibility, support organized participation, and create a foundation for future expansion.

5. Problem Statement

Event management is still handled manually or through disconnected tools by many individuals, student groups, clubs, communities, institutions, and local organizations. Organizers often use social media posts for promotion, Google Forms for registration, spreadsheets for participant lists, messaging apps for communication, and manual sheets for attendance. While these tools are familiar, they do not provide a complete event discovery and management workflow.

The major problems are described below:

- **Scattered event discovery:** Event information is often spread across social media, chat groups, notice boards, or informal messages. Participants may miss relevant events because there is no structured platform to search by location, category, and date.
- **Manual registration process:** Organizers usually collect registrations through forms or messages. This creates duplicate data, incomplete participant information, and difficulty in tracking registration status.
- **Lack of organizer approval workflow:** Many events require participant review, seat limitations, or approval before confirmation. Manual approval becomes time-consuming and difficult to track.
- **Attendance tracking problems:** Manual attendance sheets are slow, difficult to verify, and prone to mistakes. For larger events, this creates confusion and unreliable attendance records.
- **Delayed feedback and certificate process:** Feedback collection and certificate generation are often handled separately after the event. This causes delays and increases organizer workload.
- **Weak reporting and monitoring:** Organizers often lack a dashboard to view registrations, attendance, feedback, and event performance in one place.

Therefore, there is a need for a practical web-based event discovery and management platform that can support event publishing, participant registration, organizer approval, QR attendance, feedback, certificate generation, search/filter options, notifications, and dashboard-based monitoring. Eventra aims to address these problems through a realistic academic prototype with future scalability.

6. Motivation

The motivation behind Eventra comes from observing how difficult it can be to discover and manage events using disconnected tools. Many organizers still rely on social media posts, messaging apps, forms, and spreadsheets. These tools can help with small tasks, but they do not provide a complete event management system. As a result, organizers spend extra time managing registrations, confirming participants, checking attendance, collecting feedback, and preparing certificates.

The team chose this idea because event discovery and event management are common problems across many environments. Students organize seminars and club programs, communities arrange local meetups, institutions conduct workshops, and independent organizers host training sessions or social events. All of these groups need a simple and reliable platform to publish events, manage participants, and track event activity.

Eventra also has growth potential because it can begin as a local event platform and gradually expand to wider areas. A user in Barishal could publish a workshop, a club in Dhaka could manage a seminar, or a community group in another city could organize a meetup. The first version will not try to build a large enterprise platform immediately,

but it will demonstrate the foundation of a scalable event discovery and management solution. This balance makes Eventra innovative, achievable, and suitable for an academic software engineering project.

7. Expected Features and Requirements

The following features define the core functional requirements of Eventra. These features are selected to solve the main problems of event publishing, discovery, registration, attendance, feedback, certificate generation, and monitoring.

Participant Features

- **User Registration and Login:** Users can create an account and log in securely.
- **Event Discovery:** Participants can browse public events from different locations.
- **Search and Filter:** Participants can filter events by location, category, date, and keyword.
- **Participant Registration:** Participants can register for events and track their registration status.
- **QR Event Pass:** Approved participants will receive a QR-based digital pass.
- **Feedback Submission:** Participants can submit feedback after attending events.
- **Certificate Download:** Eligible participants can download generated certificates.

Organizer Features

- **Event Creation and Publishing:** Organizers can create and publish events with title, description, location, category, date, time, banner, seat limit, and rules.
- **Participant Management:** Organizers can approve, reject, or waitlist participant registrations.
- **QR Attendance Tracking:** Organizers can scan QR passes or manually mark attendance.
- **Feedback Review:** Organizers can view participant ratings and comments.
- **Certificate Generation:** Organizers can generate certificates for eligible attendees.
- **Organizer Dashboard:** Organizers can monitor total registrations, approved participants, attendance, feedback, and certificate status.

Admin Features

- **User Management:** Admins can manage users across different roles.

- **Event Monitoring:** Admins can monitor published events and platform activity.
- **Category Management:** Admins can manage event categories for consistent classification.
- **Dashboard and Reports:** Admins can view overall event statistics, user activity, feedback, and system reports.
- **Security Monitoring:** Admins can monitor suspicious activity, blocked users, and system security.

Guest Features

- Guests can view public event listings and event details.
- Guests can search events by location, category, date, and keyword.
- Guests must register or log in to register for an event.

8. Development Approach

8.1 Development Model

Eventra will be developed using the **Agile model**. This approach was selected because the project contains several connected modules, including authentication, event creation, registration, approval workflow, QR attendance, feedback, certificate generation, dashboards, and reporting. Agile will allow the team to build features progressively, test them frequently, and improve the system based on feedback.

Instead of developing the entire application in one long cycle, the team will divide the work into smaller development phases. Each phase will produce a testable part of the system. Regular review and testing will help the team identify errors early and adjust priorities when necessary. This model is suitable for an academic project because it supports learning, flexibility, and continuous improvement.

Agile Model Steps for Eventra

1. **Requirement Analysis:** Identify user roles, core features, project limitations, and functional requirements.
2. **Sprint Planning:** Divide the project into small modules such as authentication, event publishing, registration, QR attendance, feedback, certificates, and dashboards.
3. **UI/UX Design:** Prepare wireframes for homepage, event cards, event details, registration pages, organizer dashboard, and admin dashboard.
4. **Development Sprint:** Implement selected features sprint by sprint using Laravel, MySQL, HTML, CSS, JavaScript, and Bootstrap/Tailwind.
5. **Testing and Review:** Test each module after development and fix errors before moving to the next sprint.

6. **Feedback Integration:** Collect feedback from supervisor and team members, then improve usability, design, and feature flow.
7. **Final Deployment:** Prepare the final working prototype, documentation, GitHub repository, and presentation.

8.2 Tools and Technology Stack

Component	Technology
Application Type	Web application, browser-based, responsive for desktop, tablet, and mobile devices.
Frontend	HTML5, CSS3, JavaScript, Bootstrap / Tailwind CSS.
Backend	PHP with Laravel Framework, MVC architecture, routing, authentication, and role-based access control.
Database	MySQL for structured storage of users, organizers, events, registrations, attendance, feedback, certificates, and reports.
QR Code	SimpleSoftware/php-qrcode for QR event pass generation and attendance verification.
Certificate Generation	DomPDF / TCPDF for automated PDF certificate generation.
Authentication	Laravel Auth, email OTP verification, password hashing, and password recovery.
Version Control	Git with GitHub repository; all team members will be added as collaborators.
Testing	PHPUnit and Selenium for backend testing, functional testing, and UI testing.
Deployment	cPanel / AWS EC2 / BDIX hosting for standard PHP Laravel deployment.
Management	Trello / Notion / Google Meet for task tracking, sprint planning, and team coordination.
Visualization	Chart.js / Google Charts for dashboard analytics, event reports, and activity charts.
Security	SSL, Laravel Auth, role-based access, password hashing, input validation, and certificate verification security.
Design Tool	Figma / Adobe XD for wireframes, UI mockups, event cards, and dashboard prototypes.

Table 2: Eventra – Technology Stack

9. Budget Estimation

The following budget is a realistic academic project estimate for developing, deploying, and maintaining the initial version of Eventra. Most development tools are free or open-source, while the main costs are related to development effort, domain, hosting, design assets, testing, and maintenance.

Category	Details	Estimated Cost (BDT)
Development Tools	VS Code, GitHub, Composer, Laravel, MySQL tools, browser developer tools, and Postman.	Free
Frontend Development	Responsive UI, event card layout, homepage, event detail page, registration pages, organizer dashboard, admin dashboard, and public pages.	25,000 – 45,000
Backend Development	Laravel APIs, authentication, role management, event publishing, registration workflow, approval system, QR logic, certificate engine, and reports.	30,000 – 60,000
Database Design	ERD, schema design, migrations, indexing, relationships, and MySQL optimization.	10,000 – 20,000
QR and Certificate Modules	QR event pass, QR attendance scanning, certificate generation, and certificate verification.	10,000 – 25,000
Domain and SSL	Domain registration and SSL certificate for secure HTTPS access.	850 – 5,000
Hosting	BDIX hosting / cPanel hosting with suitable storage and bandwidth.	3,500 – 8,000 per year
Design and Graphics	Logo, event banner placeholders, favicon, certificate template, and UI assets.	2,000 – 10,000
Testing and QA	Functional testing, UI testing, security checking, bug fixing, and deployment testing.	10,000 – 25,000
Maintenance	Backup, minor updates, bug fixing, security monitoring, and basic support.	5,000 – 15,000 per year
Miscellaneous	Printing, documentation, presentation preparation, and contingency cost.	3,000 – 8,000
Total Estimated Cost	Overall estimated project cost for academic prototype and basic deployment.	98,350 – 2,21,000

Table 3: Budget Estimation

10. Project Timeline

The project will run over ten weeks, following the major phases of the Software Development Life Cycle. Some design, frontend, backend, and database tasks may overlap to allow parallel work and better use of team members' responsibilities.

Phase	Duration	Description
Phase 1	Week 1	Requirement analysis, problem study, scope finalization, user role identification, and task assignment.
Phase 2	Week 2	UI/UX planning, homepage design, event card design, event detail page, dashboard layout, and initial Figma mockups.
Phase 3	Week 3	Database schema design, ERD preparation, Laravel project setup, authentication planning, and GitHub repository setup.
Phase 4	Week 4–5	Backend development for users, organizers, events, participant registration, approval workflow, and role-based access.
Phase 5	Week 5–6	Frontend development and Laravel Blade integration for public pages, event discovery, registration pages, organizer dashboard, and admin dashboard.
Phase 6	Week 7	QR event pass generation, QR attendance verification, certificate generation, and certificate verification module.
Phase 7	Week 8	Feedback system, notification system, event activity reports, dashboard analytics, and event archive.
Phase 8	Week 9	Unit testing, functional testing, UI responsiveness testing, bug fixing, security checking, and performance improvement.
Phase 9	Week 10	Deployment, final documentation, GitHub organization, project report preparation, and presentation preparation.

Table 4: Project Timeline – SDLC Plan

11. Conclusion

Eventra is proposed as a scalable event discovery and management platform that can support students, clubs, communities, institutions, local organizations, and independent event organizers. The platform addresses a common problem: event discovery and event management are often scattered across social media, forms, spreadsheets, manual attendance sheets, and informal communication channels. These disconnected methods create confusion, delay, data inconsistency, and weak monitoring.

The proposed system provides a realistic academic prototype with practical features such as user registration/login, event creation and publishing, participant registration, organizer approval, QR attendance tracking, feedback collection, certificate generation, search/filter options, notifications, and organizer/admin dashboards. These features are achievable within a semester project while still representing a strong foundation for future growth.

Eventra has both academic and practical value. From an academic perspective, it gives the team experience in requirement analysis, Agile development, UI/UX design, frontend and backend implementation, database management, testing, deployment, and documentation. From a practical perspective, it can help organizers manage events more professionally and allow participants to discover suitable events more easily.

Beyond the first version, Eventra can be expanded with paid ticketing, mobile applications, AI-based recommendations, map-based search, organizer subscriptions, and multi-language support. Therefore, Eventra is not just a simple event listing website; it is a feasible, innovative, and scalable event discovery and management platform suitable for software engineering project development.